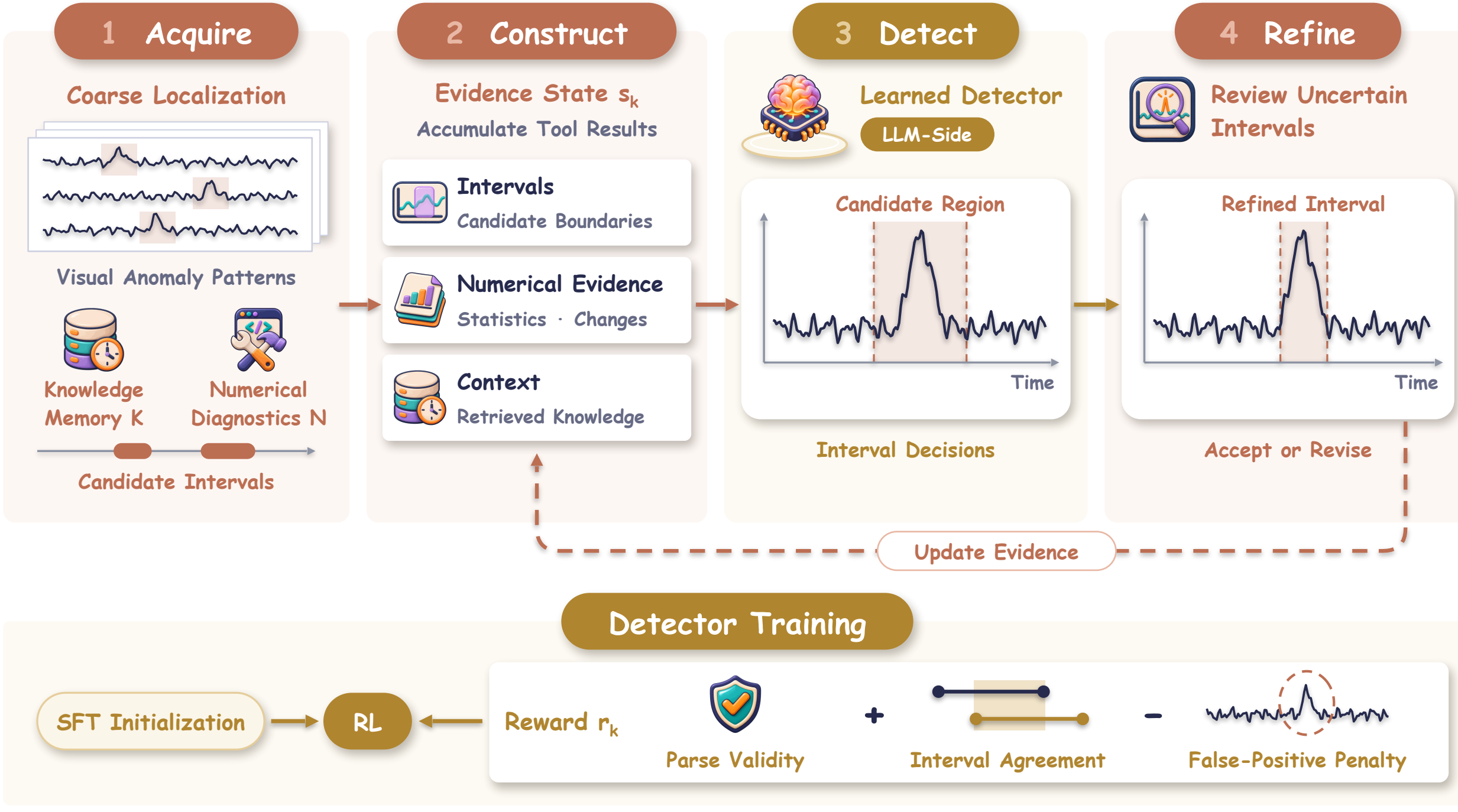

1 Acquire
Coarse Localization
Visual Anomaly Patterns
Knowledge Memory K
Numerical Diagnostics N
Candidate Intervals
2 Construct
Evidence State s_k
Accumulate Tool Results
Intervals
Candidate Boundaries
Numerical Evidence
Statistics · Changes
Context
Retrieved Knowledge
3 Detect
Learned Detector
LLM-Side
Candidate Region
Time
Interval Decisions
4 Refine
Review Uncertain Intervals
Refined Interval
Time
Accept or Revise
Update Evidence
Detector Training
SFT Initialization
RL
Reward r_k
Parse Validity
Interval Agreement
False-Positive Penalty